A080 – Units and Bookkeeping

Johann Pascher

July 2026

Contents

.1 The Separation That Everything Depends On

[K] The theory predicts ratios, not absolute values.

This sentence is the most important methodological statement of the entire block, and it cuts the error discussion into two halves that have nothing to do with each other.

A dimensionless ratio of two quantities at the same level is intrinsic: it depends on the geometry and nothing else. It needs no unit, no bridge constant, no correction factor – these cancel (A040).

An absolute value, by contrast – a mass in megaelectronvolts, a length in metres – additionally requires a conversion, and this conversion is not part of the geometry.

[B] From this follows a sharp diagnostic rule: *every deviation between theory and measurement must first be examined as to whether it sits in the geometric core or in the translation layer.* Treating both the same obscures the statement.

.2 What the Physical Constants Are Here

[STIPULATION] In the natural system of the theory, $\hbar$, c, and k_B equal one. They are not physical quantities but unit bridges.

The word bookkeeping, which appears in the corpus for this, is regularly misunderstood as dismissive. The opposite is meant. Bookkeeping here means *necessary*, not *dispensable*:

- c is the bridge between units of space and time,
- $\hbar$ the bridge between units of mass and time,
- k_B the bridge between temperature and energy.

[B] The SI separates metres, seconds, kilograms, and kelvins into distinct units. As soon as *computation* is performed in this system, the conversion factors are indispensable – without k_B an energy in kelvins would result. Ontologically they are secondary; operationally indispensable.

.3 The Four Dresses

[K] A single quantity appears in four guises:

$$m_0c^2 \;=\; E_0 \;=\; k_BT_0 \;=\; \hbar/\tilde{T}.$$

Dividing by c^2 gives a mass; by $\hbar$, a time; by k_B, a temperature. These are four expressions of the same fact, not four quantities. This identity is *exact* – it is the definition of the conversion, and it contains no approximation.

[B] E_0 itself is an empirical bridge: $E_0 = \sqrt{m_e m_\mu} \approx 7.398$ MeV *[Note 3 August 2026: the formula $\sqrt{m_e m_\mu}$ itself gives 7.348 MeV (A130, route 1); the 7.398 MeV quoted here is the geometric value from A130, route 2 ($E_0^2 = 4\sqrt{2}\,m_\mu/\xi^p$, without m_e). A130 keeps the two routes separate; the difference is 0.68 %. The number is left unchanged here, as the subsequent calculations rest on it.]* follows from measured lepton masses, not from ξ alone. The geometric derivation is in A130; the unit check in A085. A reader of only this document must know: E_0 is treated here as a given bridge constant, not as a derived quantity.

[B] Where then do deviations arise? Not in this identity. They arise where a theoretically determined dimensionless number is held against a measured quantity – and there they sit in the dress, not in the conversion.

.4 Bridge Constants

[STIPULATION] To state absolute values, the theory needs at least one quantity with dimension. This bridge constant is a declared input.

This is not a weakness of the approach, but a necessity of any theory that seeks to generate absolute numbers from dimensionless structures: from pure numbers no mass in kilograms follows. What matters solely is that the number of bridge constants is stated openly and remains small.

[B] At the same time: choosing a bridge constant is a calibration, not a test. An agreement produced by the choice of bridge must not be counted as confirmation. This rule is applied in A130 at a concrete case, where it makes the difference between a success and a self-confirmation.

.5 The Tolerance Standard in Units

[B] From the separation follows how to read a deviation. Three quantities belong together: the value, the measurement precision of the comparison quantity, and the determination tolerance of the theoretical quantity.

A deviation of one percent can be meaningless if the theoretical quantity is determined only to two places – and it can be devastating if both sides are fixed to eight places. The percentage alone says nothing. This rule is applied consistently from A100 onward.

.6 Verification Scripts

- Exactness of the four-dress identity $m_0c^2 = E_0 = k_BT_0 = \hbar/\tilde{T}$ in SI numbers, with display of the relative deviation (expected: zero up to rounding): `python/A_Serie_Skripte/a080_vier_kleidungen.py`
- Invariance of ratios under change of bridge constant, against the non-invariance of absolute values: `python/A_Serie_Skripte/a080_verhaeltnis_invarianz.py`

.7 Open Questions

First, the number of bridge constants is not tallied here. It belongs in the overall balance A230, not here – but it must appear there, otherwise the one-parameter claim of the theory is not testable.

Second, it is not shown that the bridge constant could follow from the geometry. As long as it is declared, the absolute scale remains an input. This is consistent with A020, where the same holds for the order of magnitude of ξ, and it is the same gap in a different guise.

.8 Supersedes

This document subsumes: Dok. 013 (SI), Dok. 014 (natural/SI), Dok. 015 (systematics of natural units), Dok. 122 (ratio and absolute), Dok. 134 (unit conventions), Dok. 261 (static natural units). Incorporated: P40 (ratios exact, deviations at the conversion).

.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.